

Azure ARM Template Deployment Using Custom Template

Aim

The aim of this project is to automate the deployment of a Linux-based web server in Microsoft Azure using ARM Templates (JSON), leveraging Infrastructure as Code (IaC) principles. The deployment process provisions all required cloud resources, including networking components, security configurations, and a virtual machine, while automatically installing and configuring the NGINX web server. The deployed application is made accessible through a public IP address, demonstrating efficient and repeatable cloud infrastructure automation using native Azure tools.

Project Overview

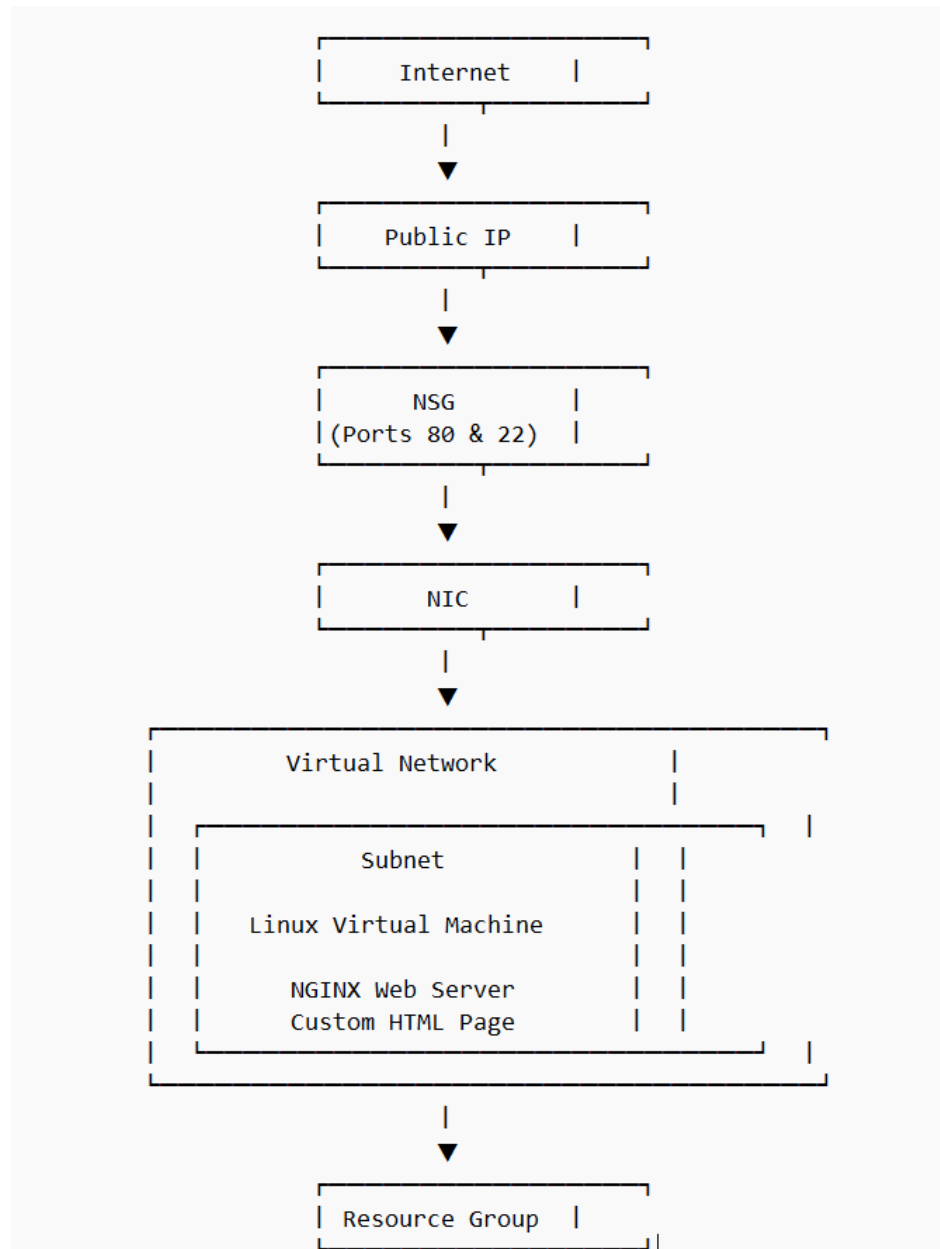
Automated Server Deployment using Native Cloud Tools is a cloud automation project that focuses on provisioning infrastructure in Microsoft Azure through ARM Templates. ARM Templates are Azure's native Infrastructure as Code (IaC) solution, enabling users to define and deploy cloud resources using JSON-based templates.

In this project, an ARM Template is used to automatically create a Resource Group, Virtual Network, Subnet, Network Security Group (NSG), Public IP Address, Network Interface, and a Linux Virtual Machine. During deployment, the virtual machine is configured with NGINX web server and a custom HTML webpage, eliminating the need for manual server setup.

The deployment is performed through Azure Portal's **Deploy a Custom Template** service, ensuring a consistent, scalable, and repeatable infrastructure deployment process. Once completed, the web server is accessible via a public IP address, allowing verification of successful deployment through a web browser.

This project demonstrates key cloud computing concepts such as Infrastructure as Code (IaC), automated provisioning, virtual networking, security management, and web server hosting using native Azure services. It highlights the benefits of automation in reducing deployment time, minimizing manual errors, and maintaining standardized infrastructure configurations across environments.

Architecture Diagram

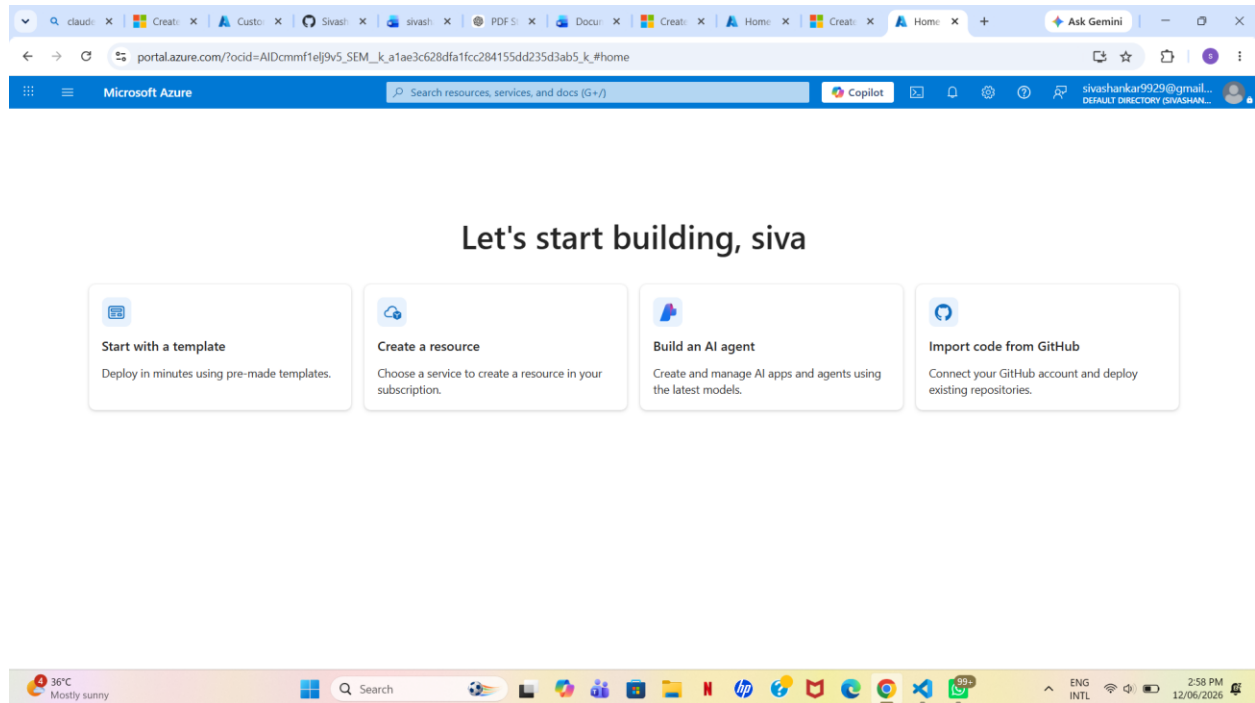


Step 1: Login to Azure Portal

Open:

<https://portal.azure.com>

Login with your Azure account.



Description:

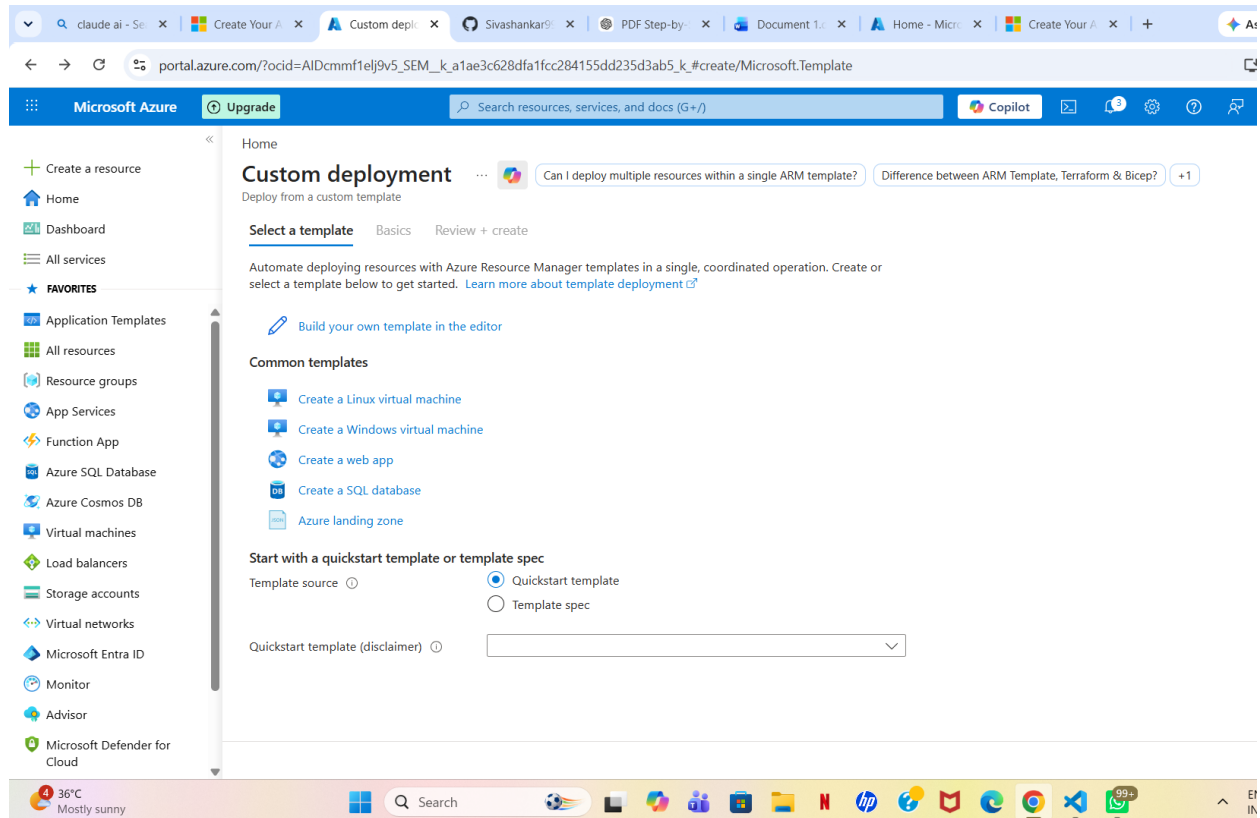
Azure Portal home page after successful login.

Step 2: Search for "Deploy a Custom Template"

In the Azure Portal search bar:

Deploy a custom template

Select **Deploy a custom template**.



Title: Deploy a Custom Template

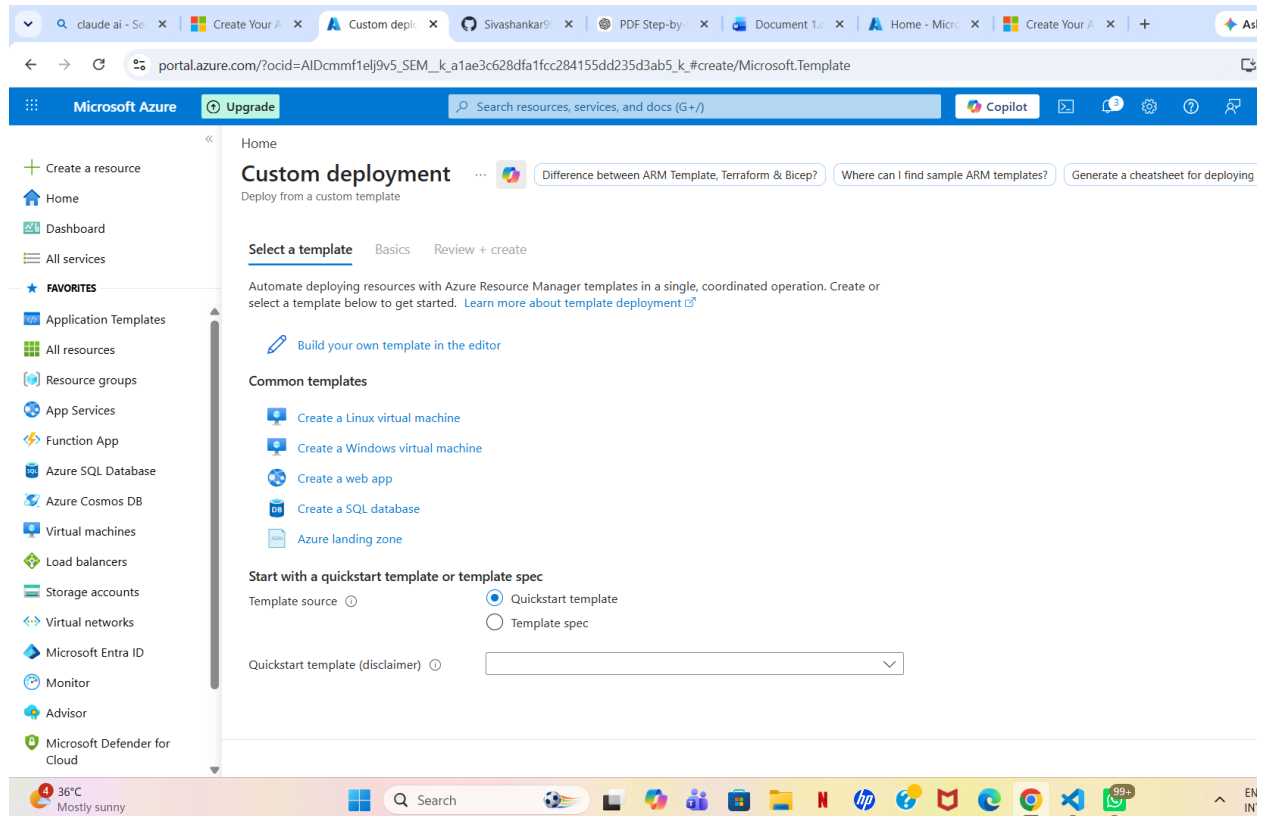
Description:

Azure Portal search results showing Deploy a Custom Template option.

Step 3: Build Your Own Template in the Editor

Click:

Build your own template in the editor



Title: Template Editor

Description:

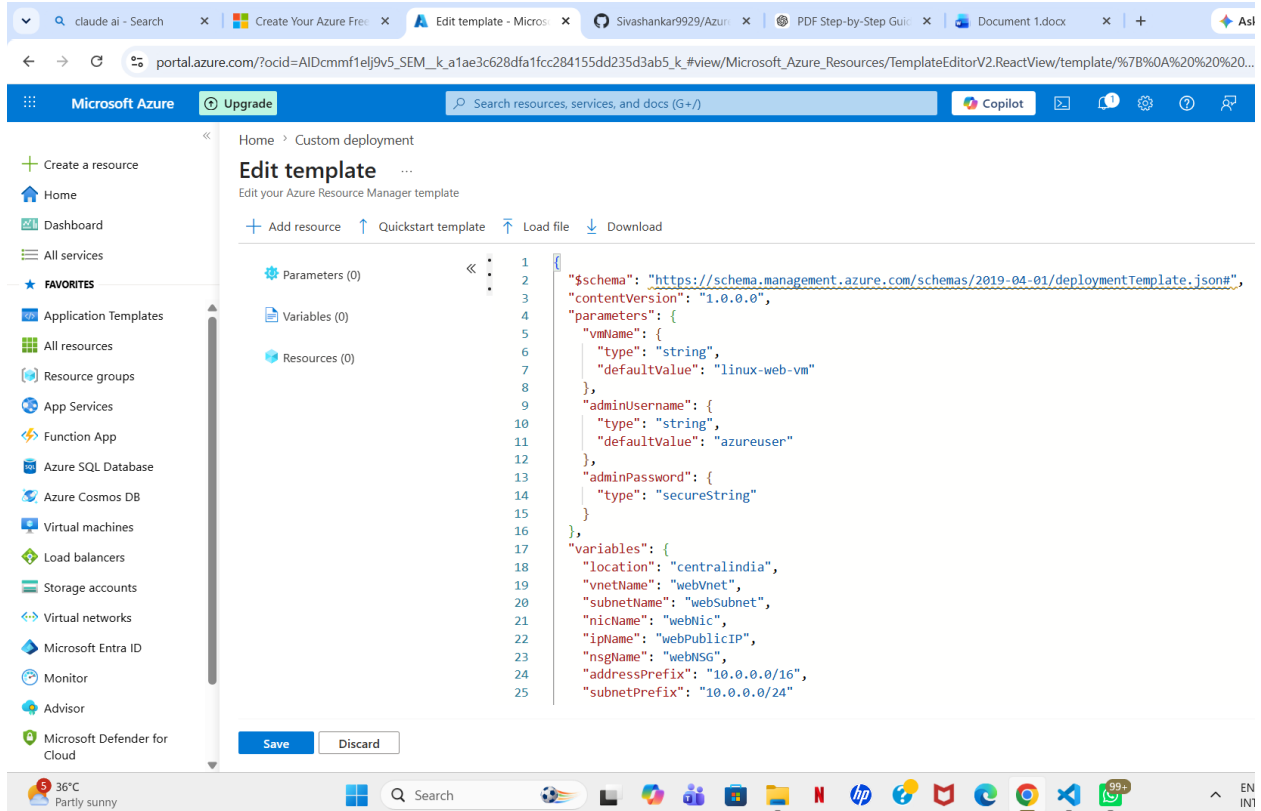
Custom Deployment editor opened in Azure Portal.

Step 4: Paste ARM Template Code

Delete the sample JSON and paste your ARM Template code.

Click:

Save



Title: ARM Template Pasted

Description:

ARM Template JSON pasted into the template editor.

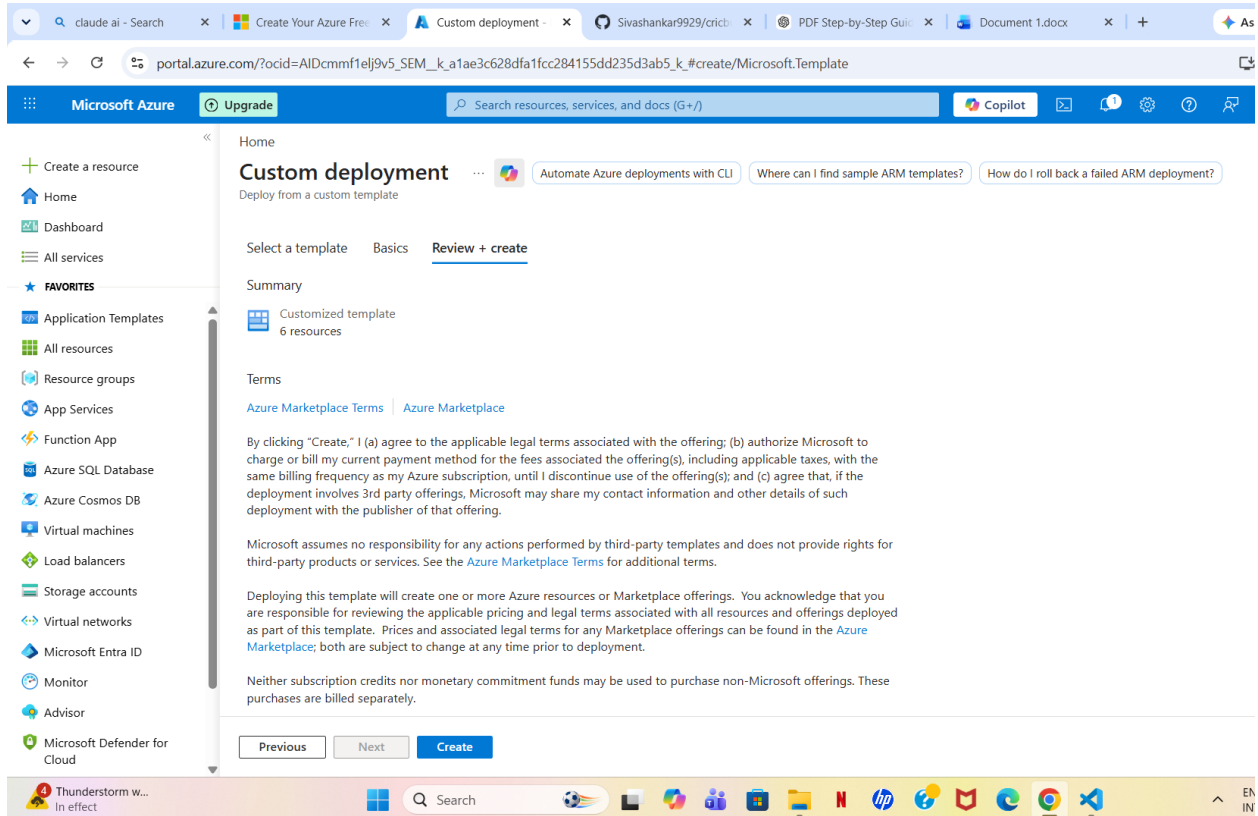
Step 5: Configure Deployment Details

Provide:

Field	Value
Subscription	Your Azure Subscription
Resource Group	Create New
Resource Group Name	arm-web-rg
Region	Central India
VM Name	linux-web-vm
Admin Username	azureuser
Admin Password	*****

Click:

Review + Create



Title: Deployment Parameters

Description:

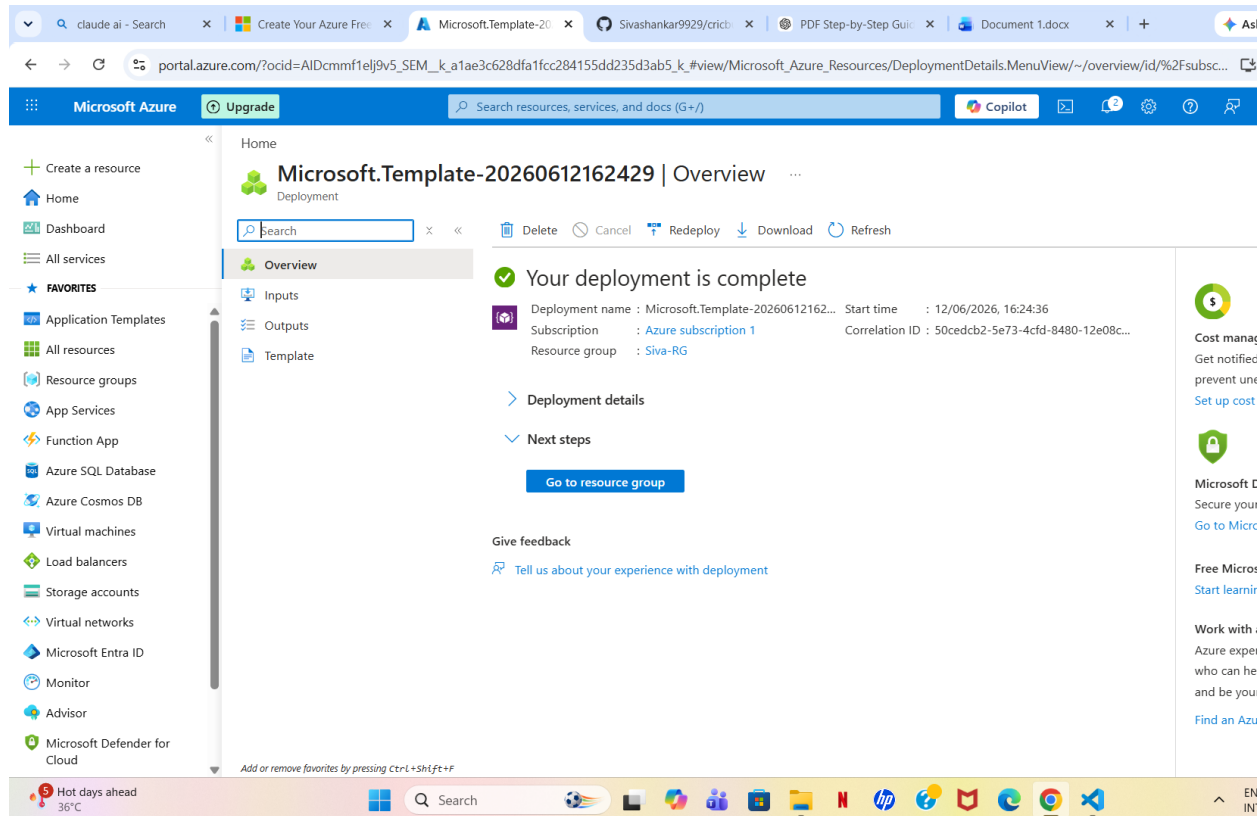
Deployment configuration page showing resource group and VM details.

Step 6: Validation Successful

Azure validates the ARM template.

Expected Message:

Validation Passed



Title: Template Validation Successful

Description:

Azure Portal validation completed successfully.

Step 7: Create Resources

Click:

Create

Deployment starts.

Screenshot Placeholder

[Insert Screenshot 7]

Title: Deployment In Progress

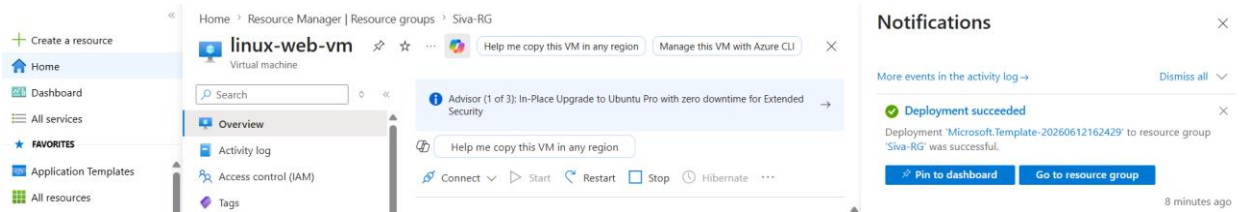
Description:

Azure deployment page showing resource creation progress.

Step 8: Deployment Completed

After a few minutes, Azure displays:

Your deployment is complete.



Description:

Azure deployment success notification.

Step 9: View Created Resources

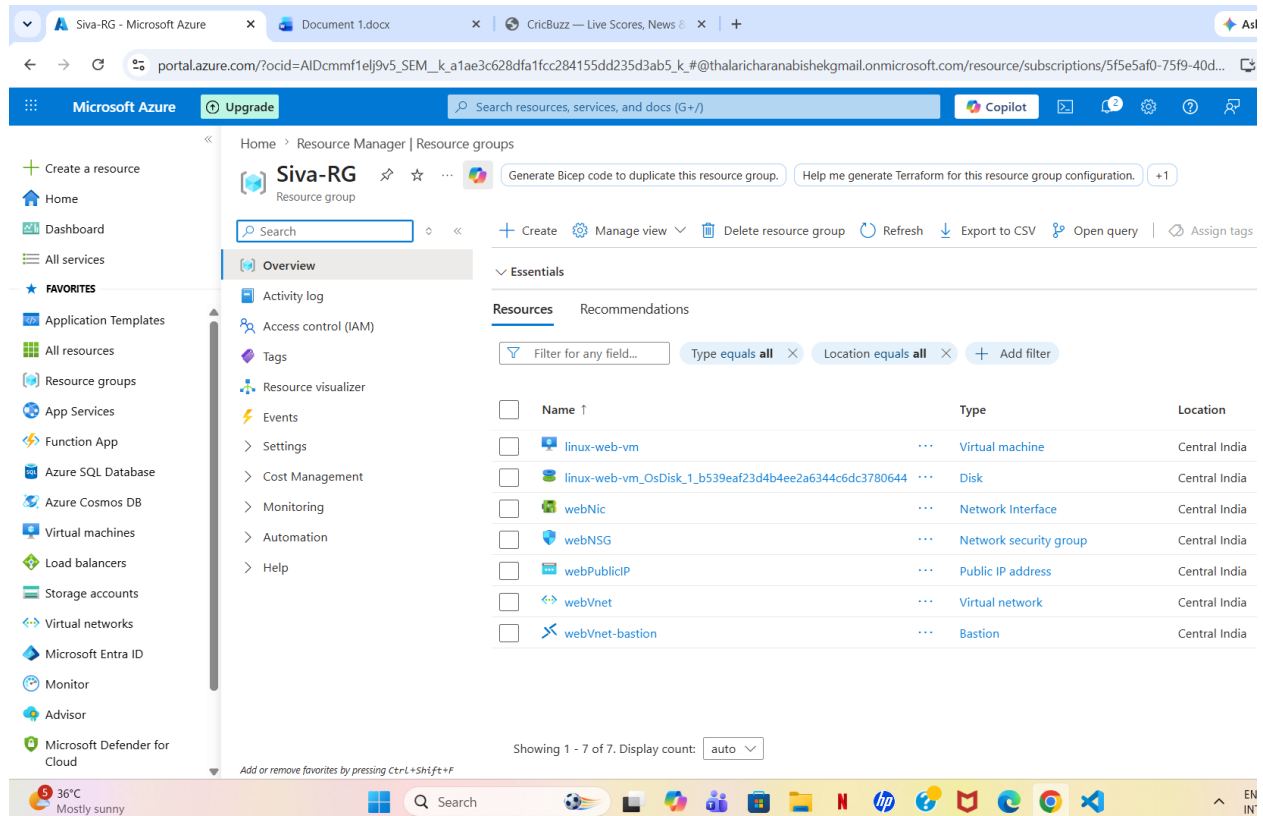
Navigate to:

Resource Groups

→ arm-web-rg

Verify:

- Linux VM
- Public IP
- Virtual Network
- NSG
- NIC



Title: Azure Resources Created

Description:

Resource Group displaying all deployed resources.

Step 10: Obtain Public IP Address

Open:

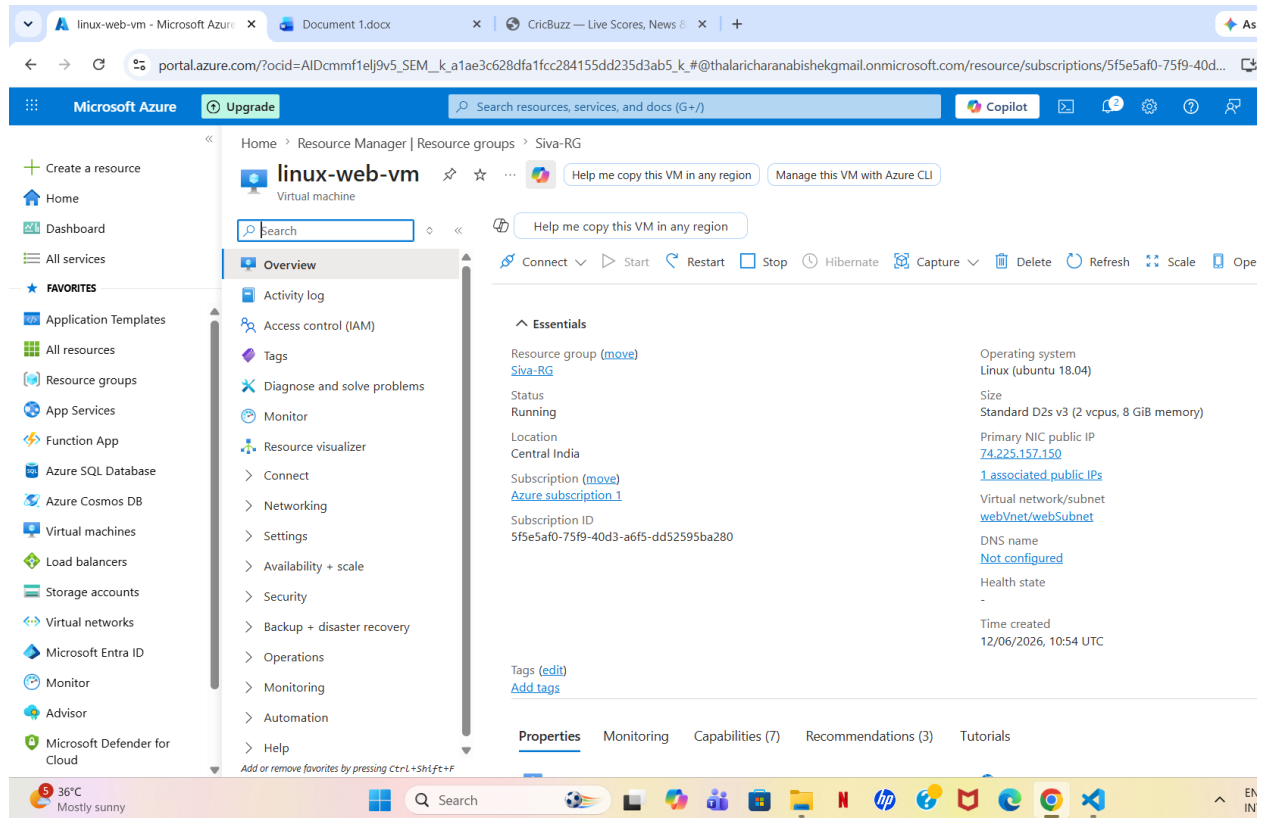
linux-web-vm

→ Networking

Copy the Public IP.

Example:

20.xxx.xxx.xxx



Title: Public IP Address

Description:

Virtual Machine overview page displaying Public IP.

Step 11: Verify Web Server

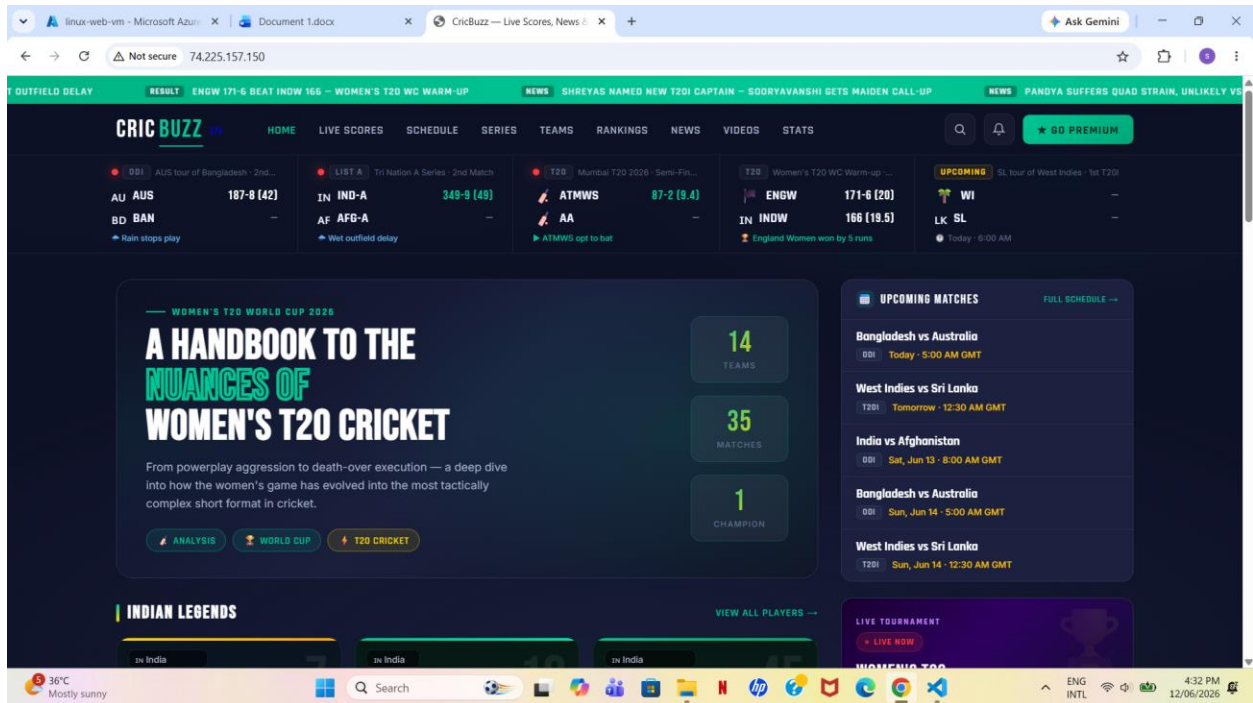
Open browser:

<http://<PUBLIC-IP>>

Example:

<http://74.225.157.150/>

Expected Result:



Title: Web Server Accessible

Description:
Browser displaying the custom webpage hosted on NGINX.

Deployment Output Table

Resource	Status
Resource Group	Created
Linux VM	Created
Public IP	Created
Virtual Network	Created
NSG	Created
NGINX Installed	Yes
Website Accessible	Yes

Conclusion

The web server was successfully deployed in Azure using an ARM Template through the Azure Portal's **Deploy a Custom Template** service. Infrastructure provisioning, network configuration, public IP assignment, and NGINX installation were automated using

Infrastructure as Code (IaC), resulting in a fully functional web server accessible over the internet.